



HAZOP Leadership training for GC

Chapter 5 - HAZOP Preparation

Technical Safety Service Division (Q-TS-TS), 1 - 3 November 2021

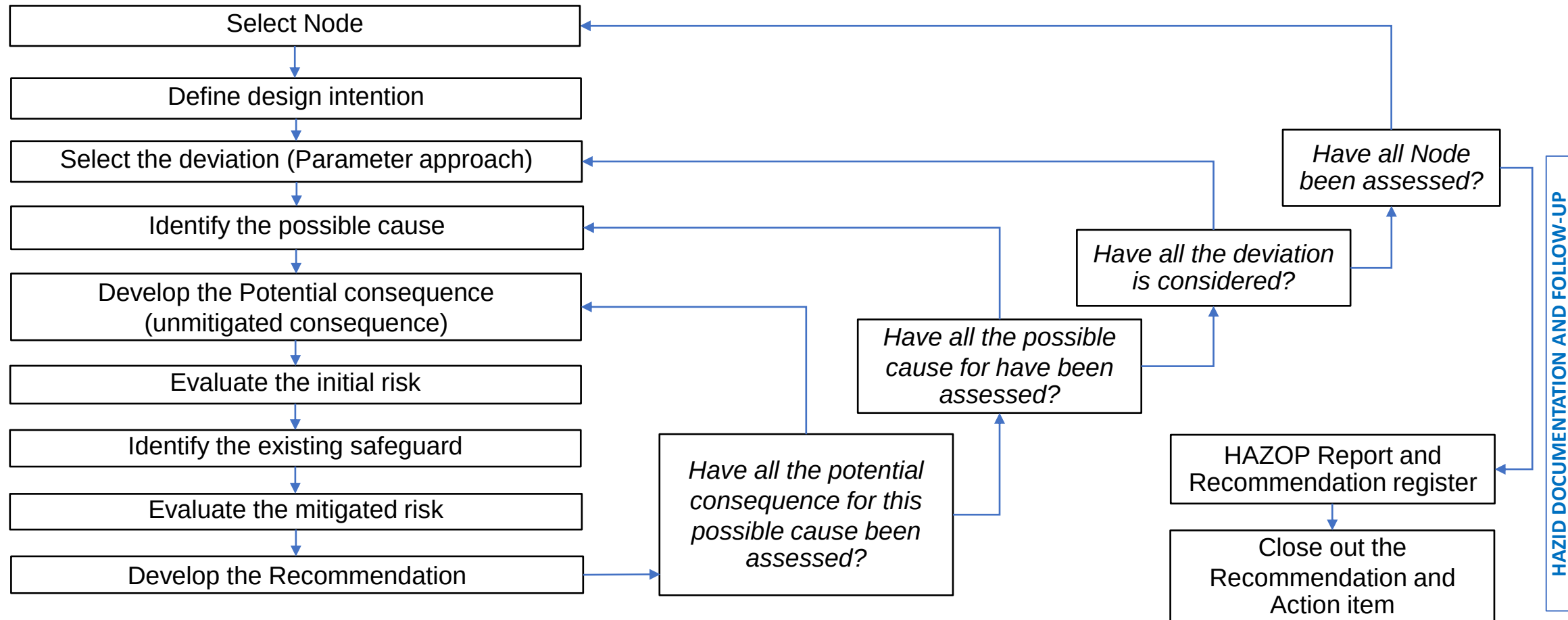
Outline of HAZOP study method

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HAZOP PREPARATION

1. Define scope & objective study
2. Planning and Scheduling
3. Team selection
4. Agreement for Risk Assessment Matrix (RAM)
5. Process safety information and other support document preparing
6. Node Preparation

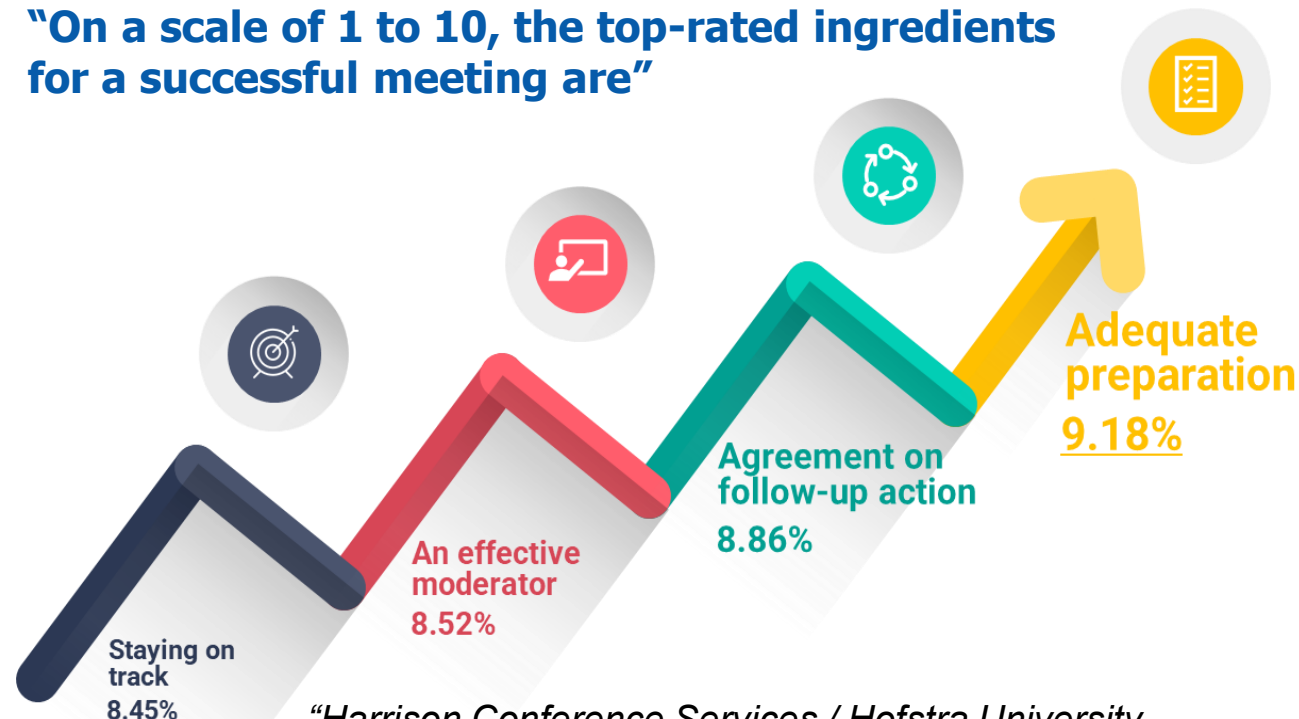
HAZOP METHODOLOGY



How Important of Preparation?

- ❑ Preparation is the key to any successful meeting
- ❑ Business leaders say one-third of the time spent in business meetings is unproductive

“On a scale of 1 to 10, the top-rated ingredients for a successful meeting are”



“Harrison Conference Services / Hofstra University survey of more than 2,000 business leaders”

Preparation Issue For HAZOP Leader

Planning

To achieve the purpose / goal of study within time and effective study

Assemble all Documents

To avoid evaluation base on incorrect information.

Select the right team

For smooth and effective brainstorming

Node Identification

For easier understanding and evaluation

Chapter 5-1 Planning

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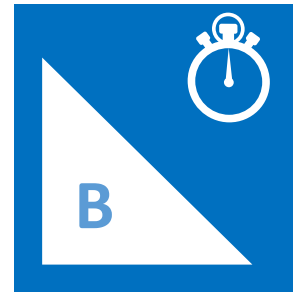
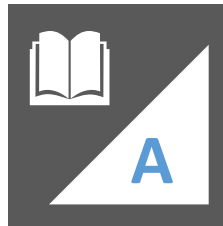
Planning

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To achieve the purpose / goal of study within time and effective study

Purpose of study

- Clear boundary of study
- New facility or modification of existing plant
- Project phase: FEED, DD, Commissioning

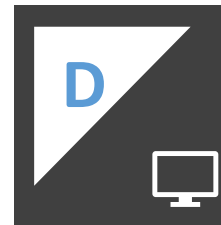


Timing

- Determine duration required of the study.
- Set up schedule

Type of Process

Continuous or Batch or Semi-batch
Automatic or manual handling



Administration

- Meeting location, room #
- Virtual meeting via MS teams

“Lead should obtain clear scope and information of change/project from HAZOP coordinator for planning”

- ❑ HAZOP team review “sessions” are each planned for half of working day (~3½ hours with breaks).

Example

MON	TUE	WEN	THR	FRI
1:00-4:30 pm	1:00- 4:30 pm		8:30am - noon	8:30am- noon

- ❑ Set day (s) of the week, starting and ending times each day.
- ❑ Estimate a long-term schedule (consider e.g., Difficulties, Project schedule constraints, Availability of team members)
- ❑ Scheduling difficulties
 - Shift schedules of available employees
 - Out-of-town participants
 - Project schedule constraints
 - Last – minute analysis

- ❑ Rules of thumb.
 - One extra session at beginning of HAZOP for:
 - Initial term review meeting
 - Team training/refreshment on methodology & ground rules.
 - Then one session per:
 - P&ID of average complexity, or
 - Unit operation/major equipment piece.
 - SIL / LOPA adds ~20% to analysis time.
 - HAZOP will require ~0.2 % of capital project cost
 - 0.25% to 0.3% if it must meet regulatory requirements
 - will vary depending on nature of project.
 - Does not include implementation of considerations
 - HAZOP will require ~1% of design cost
 - 1.3% to 1.5% if for regulatory requirements
 - More accurate than basing on capital cost.
 - Again, does not include considerations implementation

- ❑ Time requirement
 - 3 full and detailed P&IDs per day for process units
 - 4 full and detailed P&IDs per for utilities
 - more per if less busy flow sheets
 - Similar equipment, or an early stage of the project (say 4-5 per day)
 - Allow only a nominal time for duplicate trains & distribution drawing
 - In practice, 3-4 P&IDs per day is the minimum acceptable on most projects
 - Keep the pressure on from the outset

Chapter 5-2

Assemble all Documents

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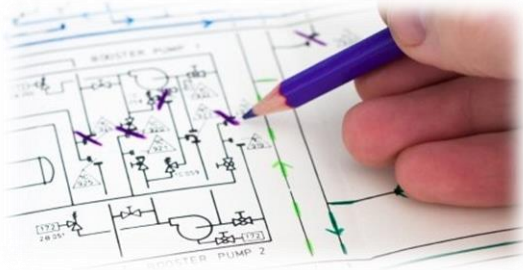
HAZOP Preparation

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2. Assemble all Documents

To avoid evaluation base on incorrect information.

- ☐ PSI and the other support information shall be listed and prepared to distribute to HAZOP Leader and team before conduct the study for reviewing prior HAZOP study
- ☐ Typical process safety information
 - Require type of document depend on the project phase
 - Operating facility not undergoing modification, PSI shall be signed off to “As-built” revision
 - Operating facility undergoing the modification or the new facility



HAZOP Leader shall evaluate the adequacy and accuracy to ensure that information have been sufficiently developed appropriate for an effective review

Caution: Garbage in, Garbage out

“What we can do when there is major PSI deficiency?”

Typical information required	Select	Define	Exceute	Operation	Non-operation
Operations and Maintenance philosophy	E	E	E	E	E
Process and Chemistry Descriptions	E	E	E	E	E
Process Control Narrative	D	E	E	E	E
Reactive Hazard (if applicable)	E	E	E	E	E
Material Safety Data Sheet	E	E	E	E	E
Plot Plan	E	E	E	E	E
General Equipment Arrangements & Elevations	-	E	E	E	E
Electrical/ Hazardous Area Classification Drawings	-	E	E	E	E
Heat and Mass balances	D	E	E	E	E
Process Flow Scheme (or PFD) and Utility Flow Scheme (UFS)	E	E	E	E	E
Process Engineering Flow Scheme and legend sheets (or P&IDs and UEFS)	D	D	E	E	E
Line lists for piping & process equipment	-	E	E	E	E
Equipment data sheet	-	D	E	E	E
Process Safeguarding Flow Scheme	D	E	E	E	E
Cause and Effect Diagrams	-	E	E	E	E
SIL Study report	-	-	E	E	E
Control valve turndown, RV, trip and alarm settings	-	E	E	E	E
Latest construction/ as-built drawings	-	-	E	E	E
Previous HAZID/HAZOP reports and Close Outs report	-	D	E	E	E
Details of change since last Process hazard analysis (PHA)	-	D	E	E	E
Other safety studies, such as Bow-tie analysis	-	-	D	E	E

E: Essential, D: Designable

Chapter 5-3

Select the right team

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Select the right team

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For smooth and effective brainstorming



Numbers too large/small

Key persons absent

Team experience

Only Some members contribute to the study.

“Leader can request required member”

Team Member	GC	IEAT PSM 2559	DIW 2543
HAZOP Leader	✓	✓	✓
Scribe	✓		
Process Engineer	✓	✓	✓
Operation representative	✓	✓	✓
Plant Safety Engineer	✓	✓	✓
Site Maintenance engineer	✓		✓
Licensor, specialist, vendor	Y/N		

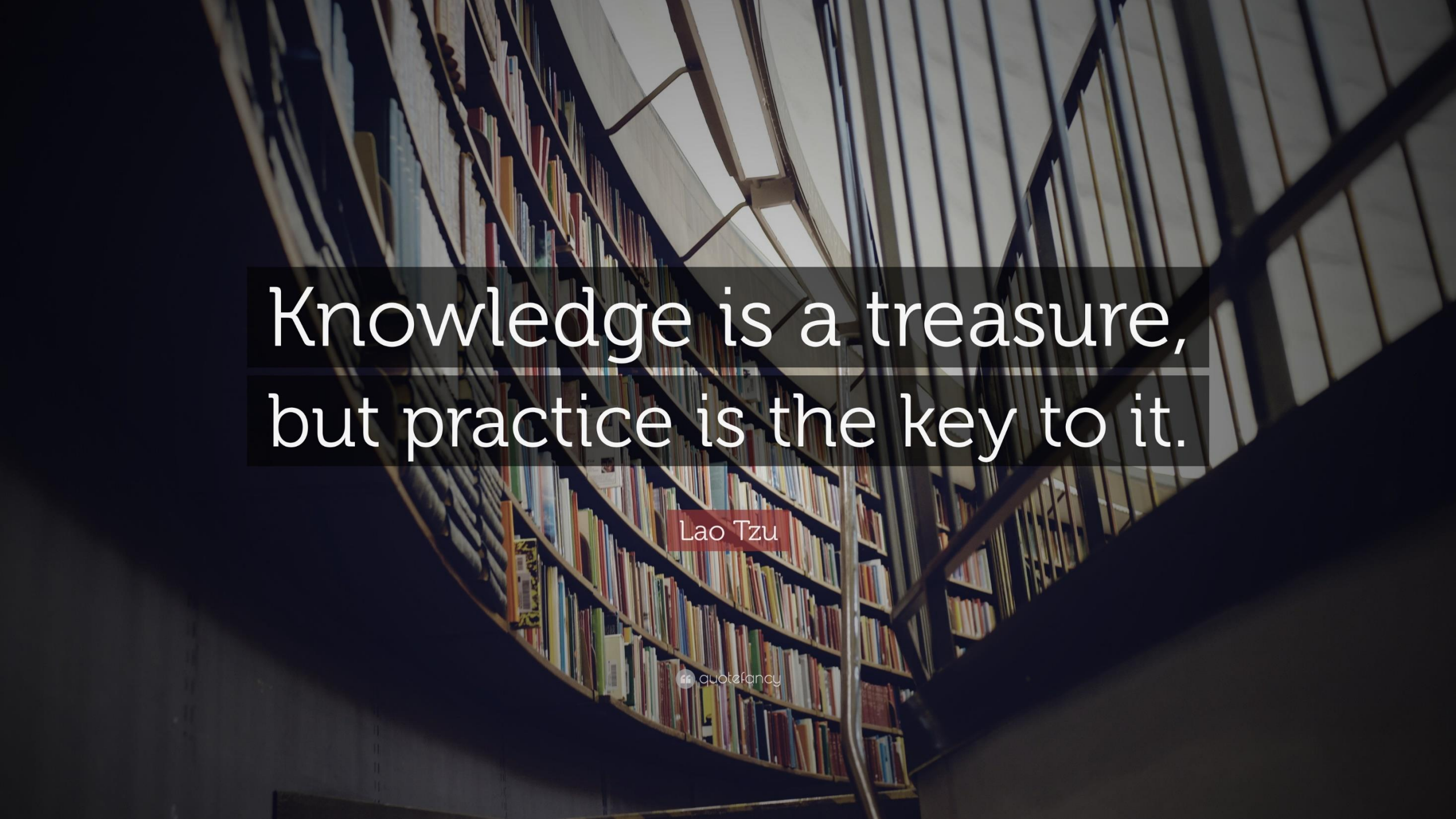
- ☐ The leader needs to have a deep understanding and considerable experience of HAZOP studies to ensure the methodology effectively and productively
- ☐ Wide experience of all the stages of HAZOP
- ☐ Extensive experience as a HAZOP study member
- ☐ Trained in the leadership of HAZOP studies
- ☐ Independence from the project with no other direct responsibility to the project leader / manager than completing the scope of the HAZOP study
- ☐ Technical competence and the ability to quickly understand the
- ☐ Meticulous attention to relevant details
- ☐ Good analytical thinking
- ☐ Motivation skills including the encouragement of creativity and open speaking. This includes to move the team towards conclusions

Responsibility of HAZOP Leader

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- ☐ Gather information for node preparation.
- ☐ Review the design to determine sufficient technical information and required skills of study team members.
- ☐ Verify adequacy of team member to ensure effectiveness of the study meeting with multi-disciplinary team
- ☐ Facilitate the meeting/workshop and ensure that all issues are covered in adequate depth.
- ☐ Being alert to time pressures and ensure that it does not compromise the quality, thoroughness, or integrity of the review.
- ☐ Advising project/site leadership of the need to delay/postpone the study until issues affecting the integrity of the HAZOP can be resolved.
- ☐ Review and approve HAZOP study report.





Knowledge is a treasure,
but practice is the key to it.

Lao Tzu

quote fancy

HAZOP Scribe Characteristic and Responsibility

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- ☐ Must be familiar with the HAZOP methodology
- ☐ Should have a technical background to ensure understanding of the team discussions without repeatedly claiming for explanations
- ☐ Following the discussion without taking part into it, filtering the items worth to be recorded and recording team in a clear diction easy to understand
- ☐ The team should always and exactly know what has been recorded
- ☐ Use a computer recording file which should be projected so all of the team can see the record
- ☐ Use a provided GC HAZOP Template



- ❑ Multi-discipline members who have experience and knowledgeable in their represented field
- ❑ Good knowledge of the design basis, design intention, operating condition and technology involved in **the operation of the process as well as the hazards of the process**
- ❑ Good knowledge of the design basis, principle & working and limitation of equipment or system involved in the operation of the process as well as current **hands-on maintenance experience in the process or system.**

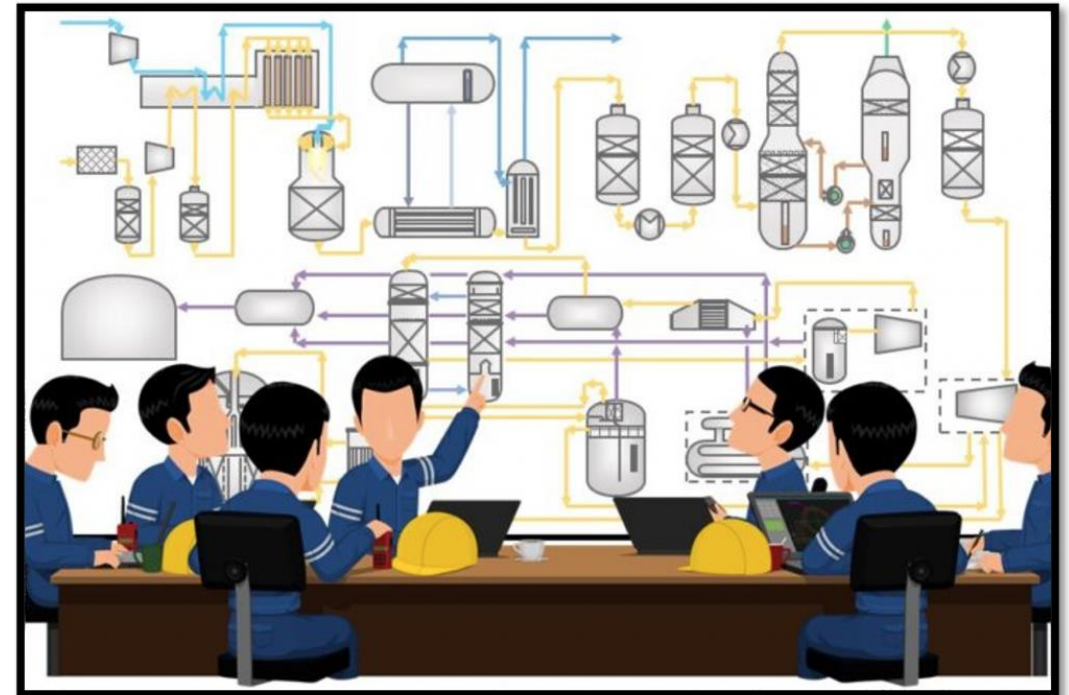
This experience involves knowing how the facilities are maintained as opposed to how they are intended to be maintained

- ❑ **Have experience with day-to-day operations and work practice.** This experience involves knowing how the facilities actually operated as opposed to how they are intended to operate and familiar with the design intent and operating, start-up, shut-down, emergency and commissioning procedures
- ❑ Good knowledge of occupational safety health and environment

Responsibility of HAZOP Team

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- ❑ **Fully participate** in the HAZOP study to their fullest ability and active (Core team member).
- ❑ Identify credible causes, develop consequences regarding hazards and operability problems including existing safeguard.
- ❑ Evaluate the risk level and suggest the recommendation for prevention and mitigation measure to reduce the risk to an acceptable level.



First HAZOP Team Meeting

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- ❑ One approach for HAZOP is to address several items before beginning the formal HAZOP methodology
 - Team qualifications
 - Documentation
 - Plant Layout / Brief Process Description / Hazards of the process
 - Previous incidents
 - HAZOP Method
 - Ground rules



Chapter 5-4

HAZOP Worksheet

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- Below are GC HAZOP worksheet

[illegible]

Working with Spreadsheets

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☐ Go e-PHA

Option 1



[MoC-PHA](#) [Revalidation-PHA/Other](#) [Project-PHA](#) [Analysis Dashboard \(MoC\)](#) [PHA Leader \(HAZOP\)](#) [Download](#) [Admin Tools](#) [Help](#)

[e-PHA LATEST NEWS](#)

Working with Spreadsheets

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❑ Go e-PHA

Option 2

SAVE

DISCARD CHANGE

EXPORT TO EXCEL

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1. MoC Information

MoC No *
MoC Title
Plant
Unit/Facility *

2. PHA Information

PHA Method *
HAZOP ☐ Attach with Complete Report (optional)
Project Stage *
FEED
Purpose *
To prevent turbine damage liquid overflow.
PHA Scope *
Identify possible hazard from add equipment P-1215B and P-1501B and relevant equipment.
Process Description *
Condensate from steam exhaust steam turbine BN-1201 and BN-1501 transfer to condensate header utilize at Dearator as Boiler feed water and to U-P1 plant.
PHA Assumption
General Note

3. Process Safety Information and Other Related Documents

FileName
P&ID Markup
Add
4. Other Study Report
FileName
NO DATA AVAILABLE
Add
5. Team Member
☐ Use External Leader

No	External Member	Role	ID	Name	Indicator	Position
1	☐	Scribe				
2	☐	MoC Champion				
3	☐	Leader				
4	☐	Member				
5	☐	Member				
6	☐	Member				
7	☐	Member				
8	☐	Member				
9	☐	Member				

Add Member Add attend date
Report Approver
CHANYANUCH K <U-P1-TE/5365>

6. Action Items

Please follow Worksheet Index, PHA Worksheet, and Recommendation to define action items. Or upload Excel Template (NOT RECOMMENDED)
SELECT FILES...
DOWNLOAD TEMPLATE

Action Items from HAZOP or other HAZARD evaluation

No. Action Detail	Node	Responsible Person	Complete with MoC Part	Due Date (Baseline)	Extend to (if necessary)	Completion Date	Approver	Approved Date	Status	Cancelled
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End of Chapter 5 : HAZOP Preparation
